AI English Spoken Tutor

Project overview: what exists, how it works, what is missing, and what to do next

Birla Open Minds International School tender (RFP-2026-AI-ANNEX-A) | Status on 29 October 2026

The project in short

A spoken English tutor for school classes (Grades Pre-Group to 8). A teacher runs a lesson on a shared smart board. A student speaks, the tutor listens, fixes the grammar and explains each mistake in simple English, and speaks the answer back. A separate admin portal lets the school manage classes, schedules and students, and see how each student is progressing and what the AI cost. The tender asks for 20 students talking at once, a reply in under 0.5 seconds, and less than $3.00 of AI cost per 45-minute class.

Where we are	Summary
Working today	A one-student-at-a-time smart board with a local tutor model, speech in and out, three lessons, and an admin backend with Classes, Schedule, Students and a per-student dashboard (38 automated tests pass).
Biggest gap	Speed and scale. Measured on a laptop, the tutor's first sound arrives about 2.1 to 2.3 seconds after a student stops talking. The tender wants 0.5 seconds, for up to 20 live voices. The board has no live audio streaming (WebRTC) yet.
Fine-tuned model	The training pipeline is built, but the fine-tuned model file is not in the project and the board runs on the base model.
Safety to-do	The admin portal has no login yet. A security review flagged this as high priority. Nothing is exposed today because it only runs on this laptop.
Dates that matter	Questions to the school: before 2 November 2026. Proposal due: 10 November 2026, 5:00 PM IST. Five-day live proof of concept for shortlisted bidders.

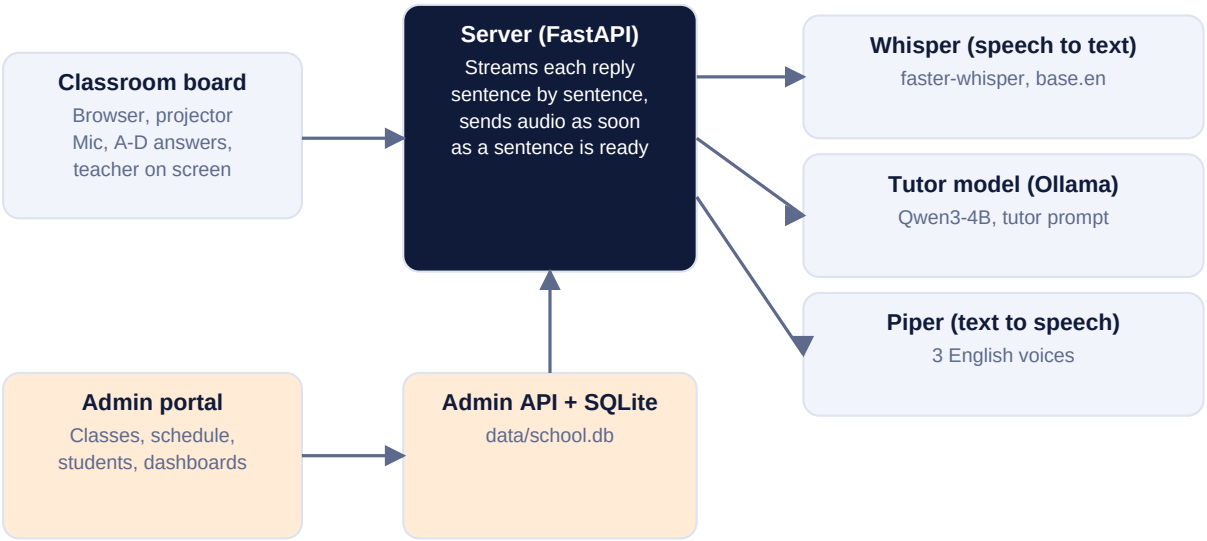
1. What the school asked for

Area	Requirement (from the RFP and its annexure)
Voice speed	End to end under 500 ms target, 850 ms ceiling, penalty above 1,000 ms. Parts: speech to text under 150 ms, first tutor word under 200 ms, first audio chunk under 150 ms.
Scale	Up to 20 students speaking at once per class. Proof of concept shows 5 live streams under 500 ms.
Cost	AI cost capped at $3.00 per 45-minute class by an automatic breaker, with a live billing dashboard (tokens, audio time, cost).
Privacy	No training on student data. Audio processed in memory and thrown away. TLS 1.3 and SRTP in transit, AES-256 at rest. Names, phone numbers and addresses removed before text goes to any model.
Safety	Stay on school topics and redirect politely. Check every reply before it is spoken. Kind, age-appropriate tone by grade. Teacher can Mute AI and Override Session in one click.
Classroom	Direct 1-on-1, hand-raise queue and round-robin turns. Live sidebar with grammar, pronunciation and vocabulary scores.
Reliability	99.5% uptime from 8 AM to 4 PM IST. Works on Chrome, Edge, Safari and Firefox with no extension.
How bids are judged	Proposal: latency design 35%, cost model 25%, WebRTC/AI experience 20%, timeline 20%. Proof of concept: live voice loop 40%, poor-network test 20%, accuracy on Indian-accent speech 20%, cost-cap audit 20%.

2. What exists today

Area	Status	Notes
Tutor model (Qwen3-4B, tutor prompt)	**Working**	Runs locally in Ollama. Replies in a fixed format: Corrected, Mistakes, Tip.
Fine-tuning pipeline	**Partial**	Dataset builder (real learner sentences), Colab notebook, test script are built. The fine-tuned model file is not in the project and the training file has only 32 examples, so it is not confirmed as trained.
Speech to text (Whisper base.en, CPU)	**Working**	About 1 second even for one sentence.
Text to speech (Piper, 3 voices)	**Working**	Fast on CPU. Piper is GPL licensed (check before selling).
Smart board	**Working**	3 lessons, A-D questions, say-it-correctly, free practice, auto-stop microphone, teacher voice that starts before the reply is finished, debug timings.
Speed work done	**Done**	Transcription now starts after 0.35 s of silence instead of waiting 0.9 s. Saved about 0.4 to 0.7 s in browser tests (two runs each, so treat as indicative).
Admin backend	**Working**	Classes, schedule (with clash detection), students, assessments, per-student dashboard. 38 automated tests pass.
Admin screens	**Partial**	Classes, Schedule, Students and Student dashboard are built. Schedule layout needs a fix and the dashboard needs a visual check.
Design mockups	**Done**	12 classroom screens and 8 admin screens as pictures, for review. Not wired to data.
Admin login	**Not started**	Needed before any real student data.
Live audio (WebRTC / LiveKit), 20 streams	**Not started**	Today the board records, uploads, then replies. This is the main tender requirement.
Hand-raise queue, round-robin, Mute AI, Override	**Not started**	Mockups only.
Live pronunciation and vocabulary scores	**Not started**	Only grammar correction text exists.
Cost limit and live cost meter	**Partial**	Per-student cost estimates exist in the admin dashboard. No live per-class limit that pauses the tutor.
Privacy and moderation layers	**Not started**	Name removal, reply checking, topic redirect, encryption at rest.
Board to admin link	**Not started**	The board does not yet send each student's result to the admin portal. Demo data is used.

3. How it works



Everything runs on one machine today, with no outside API.

The server does not wait for the whole reply. As the tutor model writes, each finished sentence is turned into speech and sent to the board at once, so the student hears the start of the answer while the rest is still being written. Short reactions such as "Spot on!" are prepared in advance and play instantly.

Two kinds of lesson step

- **Questions with fixed answers** (A-D, or say this sentence correctly): the lesson file decides right or wrong, the model only words the reaction.
- **Open sentences** (free practice or speak steps): the tutor model corrects the sentence and explains each mistake.

Admin portal

A small database (one file, `data/school.db`) holds classes, students, the weekly schedule and assessment results. The student dashboard shows this week against last week, an 8-week accuracy trend, an 8-week cost trend, a chapter-by-chapter table (attempts, accuracy, speaking minutes, cost, last taken), and the latest assessments. Accuracy is pooled over all questions, not averaged per test.

The model

Base model: Qwen3-4B Instruct. The plan is to fine-tune it (QLoRA on a free Colab GPU) on real learner sentences with human corrections from the JFLEG data set, with explanations written by a larger local model. About 20% of examples are already-correct sentences so the model learns not to invent mistakes. JFLEG is licensed for non-commercial use only, so a paid product needs the school's own data.

4. Speed, measured and estimated

Measured on this laptop (Intel Core Ultra 5 125H, no NVIDIA GPU, on charger). Three-sentence student answers, 15 test paragraphs.

Step	Time	Note
Wait for silence	0.9 s (0.35 s now)	Board setting. Transcription now starts early and is discarded if the student keeps talking.
Whisper base.en	1.0 to 1.5 s	Almost fixed: about 1.0 s for one sentence, 1.4 s for six. Flags like turning off timestamps did not help.
Tutor model, first letters	about 0.5 s	An earlier 0.17 s reading was wrong because the same sentence was cached.
Speech, first sentence	0.15 to 0.3 s	Piper on CPU.
Student stops to first sound	2.2 to 2.9 s	Test script, 3-sentence answers. In-browser test after the speed change: 2.1 to 2.3 s (before: 2.45 to 2.79 s).

Estimated (nothing below has been measured). Approximate time from a student stopping to the tutor's first word:

Setup	Estimate	Against the 0.5 s target
This laptop	2.1 to 2.3 s (measured)	4 to 5 times too slow
One GPU server, same code	0.7 to 1.0 s	Inside the 0.85 s ceiling only at the faster end
One GPU plus streaming speech recognition and 0.2 s silence	0.45 to 0.6 s	Around the target
Three GPUs, one model each, with the same tuning	0.4 to 0.55 s	Around the target, best for 20 streams
Cloud APIs (Deepgram, Haiku-class model, Cartesia)	1.0 to 1.5 s	Too slow, mostly network time

Concurrency (5 and 20 voices together) will add some delay. That can only be known from a load test on a real GPU.

5. Cost (estimates)

Setup	Per 45-minute class	Comment
One rented 24 GB GPU, class has it alone	about $0.35 to $0.85	Assumes $0.4 to $1.0 per GPU hour. Check live prices before quoting.
Three GPUs, one per model	about $1.0 to $2.3	Inside the $3.00 cap, but tight.
Cloud APIs	about $1 to $5	Wide range. 20 open streams of cloud speech recognition can exceed the cap on their own.

Idle time is the catch. A GPU costs money even when no class is running. Three GPUs kept on for 8 hours with only two classes a day would cost roughly $5 to $12 per class. Switch GPUs on only in class hours. The admin dashboard uses placeholder rates ($0.0006 per voice minute, $0.0002 per 1,000 tokens) that must be replaced with measured server costs.

6. Gaps against the tender, in priority order

#	Gap	Why it matters	Marks / risk
1	Live audio with 20 streams (LiveKit/WebRTC)	The core of the tender. Today one student at a time through upload and reply.	Proof of concept 40%
2	Speed under 0.5 s on a GPU server	Needs a GPU test, streaming speech recognition and tuning.	Proof of concept 40%, proposal 35%
3	Cost meter and automatic $3.00 breaker per class	Per-student estimates exist. A live per-class limit that pauses the tutor does not.	Proof of concept 20%
4	Accuracy on Indian-accent speech	Whisper base is weak on accents. Needs a bigger model and a test set. The base model, not the fine-tuned one, is running.	Proof of concept 20%
5	Poor-network behaviour (2 Mbps, 5% packet loss)	Untested.	Proof of concept 20%
6	Admin login and encryption at rest	Student names and results are stored. Security review flagged missing login as high.	Compliance
7	Teacher Mute AI / Override, hand-raise queue, round-robin	Designed, not built.	Compliance
8	Name removal and reply checking before speech	Mandatory safety layers, not built.	Compliance
9	Licences	Piper voices are GPL. JFLEG data is non-commercial. Both need replacing or clearing for a paid product.	Legal
10	Hosting and uptime	A free Hugging Face Space sleeps and takes 20 to 40 s per reply. 99.5% needs a proper server.	Compliance

7. What is in the project folder

Path	What it is
server.py, tutor.py, voice.py	The smart board server, the tutor model calls and prompts, speech in and out.
lessons.json, lessons.py	The lessons (3 today) and the one place that loads them, shared by board and admin.
web/ (index.html, board.js, board.css)	The classroom smart board. Early transcription was added to board.js.
admin/ (api.py, analytics.py, db.py, seed.py)	Admin backend: REST API, weekly and chapter maths and cost, SQLite storage, demo data.
web/admin/	Admin pages: Classes, Schedule, Students, Student dashboard.
tests/	38 automated tests for the admin maths and API.
build_dataset.py, finetune.ipynb, eval.py, Modelfile, data/	Model training and testing pipeline.
latency.py	Times a full spoken turn over 10 test paragraphs.
design/	Mockup pages and their screenshots (12 classroom, 8 admin), plus the text file prepared for NotebookLM.
Dockerfile, .dockerignore	One-container deploy. Updated to include the admin code.
idea.md	Parked idea: admin uploads school books and the tutor teaches only from them.
app.py	A simpler chat page for quick tests.

Nothing has been committed to git since 23 October. All the admin, design and speed work is in the working folder only.

8. How to run it

Smart board and admin portal (Ollama must be running):

```
cd "C:\Users\Aditya Sikarwar\Desktop\SLM(model)"
python -m uvicorn server:app --port 7860
```

Board: http://localhost:7860 (add ?debug to see timings). Admin: http://localhost:7860/admin/index.html
Demo data for the admin portal, the tests, and the speed test:

```
python -m admin.seed --reset
python -m pytest tests
python latency.py
```

Design mockups (pictures of every screen): run `python -m http.server 8770` inside the design folder and open http://localhost:8770/classroom/01-lesson-picker.html or /admin/login.html.

9. Suggested next steps

- **This week:** send the school your questions before 2 November (budget range, number of classrooms at once, contract length, data location, what "voice loop" is measured from).
- **Add admin login** (browser password prompt first, a sign-in page later) and finish the schedule layout fix and dashboard check.
- **Rent one GPU for a day** and rerun the speed test for 1 stream, then 5 streams. This single result decides the proposal's speed claim.
- **Link the board to the admin portal** so each finished lesson records its result.
- **Install the fine-tuned model** (or confirm it was never trained) and compare it with the base model on the 100 test sentences.
- **Plan the live audio layer** (LiveKit) and the Mute AI / hand-raise backend. Describe the books idea only as future work in the proposal.
- **Commit the work** to git once you have reviewed it.

Decisions needed from you

- Login style for the admin portal: browser password prompt now, or a full sign-in page.
- Whether the classroom design (12 screens) is approved, and which changes you want.
- Which GPU provider to test, and the budget for a one-day trial.

All speed and cost figures for GPUs and cloud services are estimates from typical performance, not measurements, and prices change. Treat them as planning numbers until tested.